Last time: informally: $y \in \mathfrak{g}$, $x \in X$.

$$\boxed{(e+y\varepsilon)\cdot x = x + \Theta_y(x)\cdot\varepsilon}$$

$G \curvearrowright X$

$$\Theta_y(x) = \frac{(e+y\varepsilon)\cdot x - x}{\varepsilon}$$

$\rightsquigarrow$ algebraic geom. f (local) function on X,

$$\rightsquigarrow \text{derivation} \quad \delta(f)(x) = \frac{f((k+y\varepsilon)^{-1}\cdot x) - f(x)}{\varepsilon}$$

"derivation"

$$\boxed{(e+y\varepsilon)^{-1}\cdot f = f + \delta(f)\varepsilon}$$

Def^n Define $\Theta : \mathfrak{g} \longrightarrow \Gamma(X, T_X)$ notation: $\Theta_y = \Theta(y)$

Let $G^{(1)} = \operatorname{Spec} \mathcal{O}(G)/I_e^2$ where I_e cuts out $e \in G$.

action of $G^{(1)}$ on structure sheaf

coaction: (locally) $\mathcal{O}_{\mathcal{U}} \longrightarrow \mathcal{O}_{\mathcal{U}} \otimes \mathcal{O}_{G^{(1)}} : \mathcal{O}_{\mathcal{U}} \otimes (k \oplus \mathfrak{g}^* \varepsilon)$

$f \longmapsto (f, \delta(f))$ $\quad \delta$ some function

$f + \delta(f)\varepsilon$

satisfy: a unit.

comultiplication: δ is derivation. $\quad T_X \subseteq \operatorname{End}_k(\mathcal{O}_X)$

coassociativity: $\Theta_{[y_1, y_2]} = [\Theta_{y_1}, \Theta_{y_2}]$ $\quad$ i.e. Θ map of Lie algebras

- Can extend Θ to map $\mathcal{U}\mathfrak{g} \to \Gamma(X, D_X)$

G-equivariant sheaves

Ex: $\mathcal{O}_X$ is a G-eq. sheaf:

$$G \times X \xrightarrow{a} X, \quad p$$

structure map: $\phi : a^* \mathcal{O}_X \longrightarrow p^* \mathcal{O}_X$

$f \in \mathcal{O}(\mathcal{U})$

$g \cdot f(-) = f(g^{-1} \cdot -) \in \mathcal{O}(g\mathcal{U})$

morphs.

fix an $g \in G$: $\mathcal{O}(g\mathcal{U}) \longrightarrow \mathcal{O}(\mathcal{U})$

$(a^* \mathcal{O}_X)_{(g, x)} = (\mathcal{O})_{g \cdot x}$

g-action is though inverse on functions

i.e. fix an g if ϕ realizes

$g^{-1} \cdot f(-) = f(g \cdot -)$.

$\mathcal{M}$ equivariant sheaf

Def^n Define $\alpha : \mathfrak{g} \longrightarrow \mathrm{End}(\mathcal{M})$ $\quad [\alpha_y = \alpha(y)]$

Restrict to $G^{(1)}$, locally.

$$\mathcal{M}(\mathcal{U}) \longrightarrow a^*(\mathcal{M})(\mathcal{U}\times G^{(1)}) \xrightarrow{\phi} p^*(\mathcal{M})(\mathcal{U}\times G^{(1)})$$

$$\mathcal{M}(\mathcal{U}) \longrightarrow \mathcal{M}(\mathcal{U}) \otimes \mathcal{O}(G^{(1)}) \xrightarrow{\phi} \mathcal{M}(\mathcal{U}) \otimes \mathcal{O}(G^{(1)}) = \mathcal{M}(\mathcal{U}) \oplus \mathcal{M}(\mathcal{U}) \otimes \mathfrak{g}^*\varepsilon \xrightarrow{\text{eval, pr}} \mathcal{M}(\mathcal{U})$$

$a^*(\mathcal{M})(\mathcal{U}\times G^{(1)}) \cong \mathcal{M}(\mathcal{U}) \otimes \mathcal{O}(G^{(1)})$, $\quad a^{-1}(\mathcal{U}) = \mathcal{U}$

$\mathcal{M}(\mathcal{U}) \longrightarrow a^*\mathcal{M}(a^{-1}(\mathcal{U}))$

$$\boxed{\mathcal{M}(\mathcal{U}) \longrightarrow a_*a^*\mathcal{M}(\mathcal{U})}$$

Associativity: α is a map of Lie algebras

$\mathcal{O}$-linearity: $[\alpha_y, f]\cdot m = \langle \theta_y, df\rangle \cdot m$

Remarks: ① Locally, $\alpha_y(m)$ given by formula:

$$\alpha_y(m) = \frac{(e+y\varepsilon)^{-1}\cdot m - m}{\varepsilon}$$

$$\alpha_y(fm) = \frac{(e+y\varepsilon)^{-1}\cdot(fm) - fm}{\varepsilon}$$

$$= \frac{\left((e+y\varepsilon)^{-1}\cdot f \times (e+y\varepsilon)^{-1}\cdot m\right) - fm}{\varepsilon}$$

$$= (y\cdot f)m + f(y\cdot m)$$

$$\alpha_y(fm) - f\alpha_y(m) = \theta_y(f)\, m$$

$g\cdot(fm) = (g\cdot f)(g\cdot m)$

$\mathcal{M} \quad \mathbb{R}\text{-mod}$

$\cup \quad \cup$

$\mathcal{O} \quad G$

$$\boxed{\begin{array}{l}\text{Exercise: } \mathcal{M} = \mathcal{O}_X \\ \alpha_y = \theta_y\end{array}}$$

Remark: Not rigorous ... not allowed to expand $(e+y\varepsilon)$

Can make it rigorous using "product rule trick"

$$\begin{tikzcd} \mathcal{O}_{g\cdot x} \arrow[r, "\phi"] \arrow[d] & \mathcal{O}_{g\cdot x} \arrow[d] \\ \mathcal{O}_x \arrow[r, "g\cdot\phi"'] & \mathcal{O}_x \end{tikzcd}$$

Examples:

① $\mathcal{M} = \mathcal{O}_X$, $\quad \alpha_x = \theta_x$

② $\mathcal{M} = T_X$ $\quad \alpha_x(\theta) = [\theta_x, \theta]$ $\quad$ Why?

③ $\mathcal{M} = D_X$ $\quad \alpha_x(P) = [\theta_x, P]$

Prop: $\mathcal{M}$ is a D_X-module. $\quad$ Let $\theta_y : \mathfrak{g} \longrightarrow T_X(X) \hookrightarrow D_X(X) \hookrightarrow \mathrm{End}(\mathcal{M})$

$\alpha_y : \mathfrak{g} \longrightarrow \mathrm{End}(\mathcal{M})$

$$\boxed{\eta_y = \theta_y - \alpha_y}$$

$\eta : \mathfrak{g} \to \mathrm{End}_k(\mathcal{M}) \supseteq \mathrm{End}_{D_X}(\mathcal{M})$

Lemma: η_y is D_X-linear

Ex: $\mathcal{M} = D_X$: $\quad \eta_y(P) = \theta_y(P) - \alpha_y(P) = \cancel{\theta_y P} - (\cancel{\theta_y P} - P\theta_y) = P\theta_y$.

Def The Beilinson–Bernstein localization functor. $\mathcal{U}_0\mathfrak{g} = \mathcal{U}\mathfrak{g} \otimes_{\mathfrak{Z}\mathfrak{g}} \mathbb{C}$

$$\mathrm{Loc}: \mathrm{Mod}(\mathcal{U}_0\mathfrak{g}) \xrightarrow{\sim} \mathrm{DMod}(G/B)$$

$$M \longmapsto \mathcal{D}_X \otimes_{\Gamma(\mathcal{D}_X)} M \qquad \Gamma(\mathcal{D}_X) \simeq \mathcal{U}_0\mathfrak{g}$$

Thm $\Gamma: \mathrm{DMod}^{\mathcal{U}}_c(G/B) \longrightarrow \mathcal{O}'_0$ = $\mathfrak{Z}\mathfrak{g}$ acts by 0 ← struct

↖ holonomic b/c orbits finite

→ $\mathfrak{n}$-locally finite

v. fg.

no fg semisimplicity

Proof: coherent ⟷ fg.

abstract nonsense. Loc compact ⟷ $\mathrm{Hom}(X, -)$ commute w/ all

Brian Lino Hunter, alg geom

Key fact: exp. $\mathfrak{n} \xrightarrow{\wedge} \mathcal{U}$

If $M = \Gamma(\mathcal{M})$ ← stay $\mathfrak{n}$-q, $\mathcal{U}$-shift $\Rightarrow \mathcal{M}$ is partial $\mathfrak{n}$-rep.

$\Rightarrow \mathcal{M}$ is locally $\mathcal{U}$-finite

$\overset{\text{exp}}{\Longrightarrow} \mathcal{M}$ is locally $\mathfrak{n}$-finite

If M is $\mathcal{U}_0\mathfrak{g}$-mod, l.f.

$\mathcal{U} \circlearrowright \mathrm{Loc}(M) = \mathcal{D}_X \otimes_{\mathcal{U}_0\mathfrak{g}} M \circlearrowleft \mathcal{U}$, b/c $\mathfrak{n}$-l.f.

$\circlearrowleft \mathcal{U}$

Sat: $\mathcal{U}$ acts on $\mathcal{D}_X \otimes M$

$u \cdot (P \otimes m)$: $uP \otimes m + P \otimes u \cdot m$

Lemma: Show that acts on $\mathcal{D}_X \otimes_{\mathcal{U}_0\mathfrak{g}} M$

Proof: Start eq $\eta_y(P \otimes m) = 0$.